DT081A - Signal and Image Processing
Exercise 1 Solutions

Question 1. Define mathematically a non-damped mechanical signal that oscillates around zero with amplitude of 20 cm, if it starts at 0 at time t = 0 and oscillates four periods.

Answer: The resulting function is

$$x : [0s, 4s] \to [-20cm, 20cm] \tag{1}$$
$$t \mapsto 20\text{cm} \cdot \sin(2\pi t s^{-1}). \tag{2}$$

Question 2. Define mathematically a pure sinusoidal audio signal x(t) that consists of 100 Hz, with zero phase-shift with an exponentially declining amplitude from 2 with a rate of $\sigma = 4$.

Answer: The function in question is

$$x(t) = 2 \cdot e^{-4t} \sin(2\pi 100 t s^{-1}). \tag{3}$$

Question 3. Formulate the mathematical description of the above audio signal being sampled at a sampling rate of $f_s = 1000$Hz, i.e. $t = n \cdot T_s$, where $T_s = \frac{1}{f_s}$.

Answer: The sampling can be written as

$$S = \{x(n \cdot T_s) \mid n \in \mathbb{N}, T_s = \frac{1}{1000}\text{s}\}. \tag{4}$$

Question 4. A digitized version of the signal in Question 1 can use 8 bits for the whole range to cover the signal amplitude values. Write the possible values of the signal amplitude.

Answer: Since 8 bits can differentiate at best $2^8 = 256$ chunks of information, the possible values of the signal amplitude are evenly spaced within the signal's range, meaning we get

$$B = \left\{ -20\text{cm} + z \cdot \frac{40}{255}\text{cm} \mid z = 0, 1, \ldots, 255 \right\} \tag{5}$$

Question 5. A *digital* signal processing algorithm filters a signal $s_n, n = 0, \ldots$ with a filter g. Write the mathematical description and support Ω.

Answer: The resulting signal is

$$(s \star g)_n = \sum_{i=-\infty}^{\infty} s_i \cdot g_{n-i} \tag{6}$$

and the support is

$$\Omega = \{n \in \mathbb{Z} \mid (s \star g)_n \neq 0\}\,. \tag{7}$$

Question 6. Describe the difference between audio processing and image processing with respect to dimensions, contents, and support.

Answer: Audio signals are 1-dimensional time-signals, while images have two dimensions (height and width). The content of an audio signal is amplitude of a sound-wave with respect to time, while the content of an image is intensity with respect to spatial coordinates. The support of an audio signal is usually a time-interval sans the times when there the signal is zero. The support of an image signal is a square sans the regions where the color intensity is zero.

Question 7. Give examples of different Image Processing and where it could be applied.

Answer: There are many different image-processing approaches, but here are some of them:

Low-pass filtering: If an image has a large amount of noise - be it Gaussian, Poisson or salt-and-pepper -, a low-pass filter can often mitigate the noise (denoising).

Super-resolution: The concept of increasing the resolution of an image by any means. This can be very useful to improve images taken from far-away or with low-quality cameras. Neural networks are an excellent tool for this, be it as a full black-box approach, but more optimally using approaches like unrolled or plug-and-play (or generative models for extreme levels of super-resolution).

Inpainting: This refers to the filling-in of missing parts in an image. There are many different approaches to this, but neural-networks appear to deliver the best results (e.g. diffusion models).

Deblurring: The attempt to remove some kind of filter that has been applied to an image. This can be used to remove atmospherical blur, defocus blur or motion blur. Once again, neural networks have emerged as superior tools for this purpose (e.g. plug-and-play approaches, or unrolled).

Question 8. What is the reason for using Red-Green-Blue representations of colour images?

Answer: Red-Green-Blue (RGB) representations have been chosen because our eyes also have cones for those three colors, meaning this mirrors our own perception of colors. This model furthermore has the advantage that decomposing colors into these three components (and their intensities) allows further separation of signals.

Question 9. Give possible applications of the examples given in the lecture for

- Video processing
- Light field processing
- Plenoptic processing

Answer: **Video Processing:**

- **Filter:** Filters can be applied to reduce noise (denoising) in a Video or enhance edges. Furthermore, they can be used to stabilize shaky videos.
- **Inpainting** can be used to fill in missing parts in a video or remove occlusions (e.g. a finger in front of the lens).
- **Spatio-temporal super-resolution** can be split up into two different goals: Spatial super-resolution aims to increase the resolution of each frame, which may be desirable if the camera used was not very good or one wants to zoom in on a part of the video. Temporal super-resolution increases the frame-rate by interpolating frames that were not captured by the camera.

- **Deblurring** can be quite useful when the camera is moved during capture and this results in motion-blur.
- **Denoising** is essential when there is high sensor-noise, which is likely to happen for high frame-rate videos or when videos are captured in very dark environments.

Light field processing:

- **View generation** can be used to generate a whole light-field from a single view or to generate missing views in a captured light-field
- **Refocusing** can be important if the camera was focused on one part of the scene but one wants another part of the scene (or the entire scene) to be in focus.
- **Zooming** is the application of super-resolution to a specific spatial part of a light-field. This can be useful to allow for navigation in a scene (the observer moving closer to the scene).
- **Filtering** can be used on the epipolar-plane-image (EPI) to improve view-consistency.
- **Inpainting** is particularly useful when trying to remove occlusions or create novel views that would include previously unseen information. This can be crucial for view extrapolation.
- **Spatio-angular super-resolution** can be useful to deal with the spatial-angular resolution trade-off encountered with many light-field capturing solutions. If lenses are used to create many sub-aperture images, one trades angular resolution for spatial resolution. Here, super-resolution can mitigate this trade-off.
- **Deblurring** while light-field deblurring (on e.g. motion-blur) has similarities to 2D image deblurring, in a light-field one has extra information stemming from different views. This allows for higher quality deblurring but also imposes the problem of view-consistency. Deblurring each image separately may lead to inconsistencies between views that deteriorate overall light-field quality.
- **Denoising** is an important topic for corner views generated by lenslet cameras. These cameras tend to have (very) dark corner views which leads to them often being discarded. A good denoising approach could mitigate this issue.
- **Depth extraction** is useful when trying to ascertain geometry from captured images. This can be helpful when trying to create a 3-D representation of a scene or object allowing for digitization or other further use.

Plenoptic processing Plenoptic processing includes light-field processing, but is a more general concept represented by the plenoptic function. This said, the examples for light-field processing also hold here.

Question 10. Describe the difference between Image Processing and Computer Vision.

Answer: Image processing refers to modifying an image (i.e. input and output are images). Computer vision refers to the field of interpreting and categorizing data in images. This can include image processing approaches but can also extend to classifying images, finding objects in images or generating images from scratch.

Question 11. What is the reason for introducing machine and deep learning algorithms instead of the classical image processing methods?

Answer: Machine and deep learning algorithms are powerful for extracting complex patterns and features from data. Classical image processing methods, on the other hand, rely on hand-crafted rules

and filters, while machine learning methods can automatically learn representations and relationships from large datasets. This allows for better generalization to diverse tasks and improved performance, especially in situations with large amounts of data. An example is plug-and-play methods, where knowledge of image degradation (e.g., blurring) is combined with trained neural networks to reconstruct high-quality images.

Question 12. Based on the above answer, why should you also understand classical approaches for Image Processing?

Answer: As already mentioned above, it appears that some of the best solutions can be found by using both: classical approaches as well as machine learning. If an explicit or classical solution is available for a given problem, it is often preferable to use it (as this needs to trainable parameters and can be applied without pre-training). If, on the other hand, large amounts of data are available, a trained machine learning method can improve results. Many state-of-the-art solutions combine the use of neural networks with classical theory (e.g., diffusion models, plug-and-play algorithms, deep equilibrium models, unrolled methods ...).

Question 13. Consider slide 33 of Lecture 1. What kind of processing do you envision at the different processing points, and what would the dimensionality of the data be?

Answer: **Pre-processing and compression** can include things like simultaneous localisation and mapping (SLAM) as well as image processing (like denoising, deblurring, edge detection or normalization). Compression is done to improve data-efficiency, removing redundant or superfluous data. The input-data can be of different dimensions depending on the input (2D for pure images, 3D for video, 4D for video plus sound ...). The pre-processed data can be of a lower dimension but may also have a higher dimension (e.g., compression via autoencoder).

Post-processing then de-compresses the images to allow for further use. The output of this step needs to be in line with the renderer used. This step needs to extract all the information available in the compressed data and enrich it as much as possible for the rendering step.

Question 14. Two complex numbers are $c_1 = -3 - j5$ and $c_2 = 5 + j3$.

(a) Compute their amplitudes.

(b) Compute their phases/arguments.

(c) Express them in polar form.

(d) Draw them and their complex conjugate in the complex plane. Indicate amplitude and phase for each of these vectors.

Answer: The amplitudes (see lecture 1 slide 41) are

$$|c_1|^2 = (-3)^2 + (-5)^2 \tag{8}$$
$$= 34 \tag{9}$$
$$\approx 5.81 \tag{10}$$

and

$$|c_2|^2 = 5^2 + 3^2 \tag{11}$$
$$= 34 \tag{12}$$
$$\approx 5.81. \tag{13}$$

Answer: Their phases/arguments (see lecture 1 slide 41) are

$$\varphi_1 = \arctan\left(\frac{5}{3}\right) + \frac{\pi}{2}\left(1 - \text{sign}(a)\right) \tag{14}$$
$$\approx 1.03 + \pi \tag{15}$$
$$\approx 4.17, \tag{16}$$

and

$$\varphi_2 = \arctan\left(\frac{3}{5}\right) + \frac{\pi}{2}\left(1 - \text{sign}(a)\right) \tag{17}$$
$$\approx 0.54 + 0 \tag{18}$$
$$\approx 0.54. \tag{19}$$

Answer: The polar forms are

$$c_1 = \sqrt{34}\cos(\phi_1) + j\sqrt{34}\sin(\phi_1) \tag{20}$$
$$\approx 5.81(\cos(4.17) + j\sqrt{34}\sin(4.17)), \tag{21}$$

and

$$c_1 = \sqrt{34}\cos(\phi_2) + j\sqrt{34}\sin(\phi_2) \tag{22}$$
$$\approx 5.81(\cos(0.54) + j\sin(0.54)). \tag{23}$$

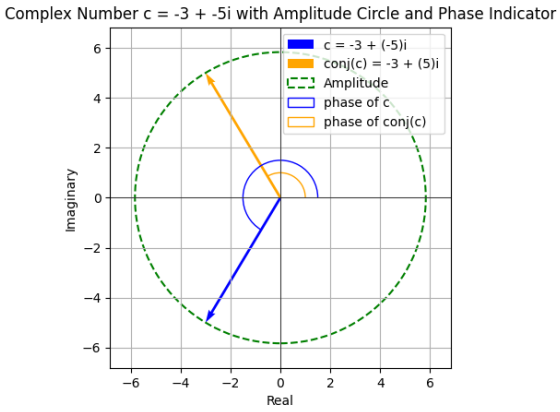


Figure 1: Plot for c_1

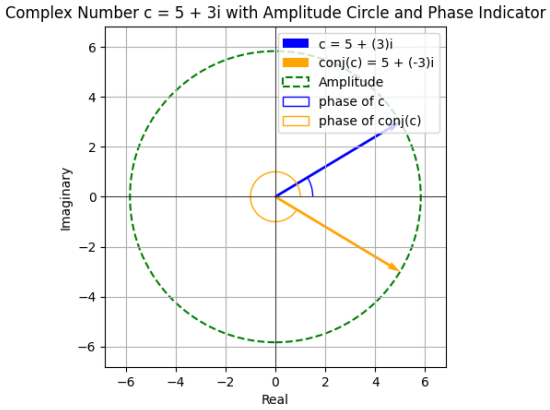


Figure 2: Plot for c_2

Question 15. A time-discrete signal $s(n)$ consists of the following sequence: $2, 8, 3, -2, -1, 7$ for $n = [0, 5]$. Write the signal in vector form.

Answer: In column form, the vector is

$$s = [2, 8, 3, -2, -1, 7]^T. \tag{24}$$

Question 16. The $s(n)$ above is filtered by g with the convolution $y(n) = \sum_{k=0}^{2} g_{n-k} s(k)$, where $g_k = 1, -2, 2$ for $k = 0, 1, 2$. Write the convolution as a matrix multiplication for $y(n), n = [0, 5]$.

Answer: To define this as a matrix multiplication, we first assume $g_k = 0$ for $k \notin 0, 1, 2$. This means we get

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & -2 & 1 & 0 & 0 & 0 \\ 0 & 2 & -2 & 1 & 0 & 0 \\ 0 & 0 & 2 & -2 & 1 & 0 \\ 0 & 0 & 0 & 2 & -2 & 1 \end{bmatrix} \tag{25}$$

and thus

$$y = Gs. \tag{26}$$

Question 17. A function is $x(t) = e^{-100t} u(t)$, where

$$u(t) = \begin{cases} 1 & \text{if } t \geq 0, \\ 0 & \text{if } t < 0 \end{cases}$$

is the heavy-side function (= step function).

(a) Find the complex coefficients with the Hilbert space

$$V : \Phi = \left\{ \varphi(t) = e^{j\omega t} \mid \omega, t \in \mathbb{R} \right\}.$$

(b) What is another name for this representation?

Answer: To find the complex coefficients within the given hilbert space, we need to calculate the inner product

$$\begin{aligned} X(\omega) &= \langle x(\omega), e^{j\omega t} \rangle && (27)\\ &= \int_{-\infty}^{\infty} x(t) e^{-j\omega t} \mathrm{d}t && (28)\\ &= \int_{-\infty}^{\infty} e^{-(100+j\omega)t} u(t) \mathrm{d}t && (29)\\ &= \int_{0}^{\infty} e^{-(100+j\omega)t} \mathrm{d}t && (30)\\ &= \frac{-1}{100+j\omega} (\lim_{h\to\infty} e^{-(100+j\omega)h} - 1) && (31)\\ &= \frac{1}{100+j\omega} && (32)\\ &= \frac{100-j\omega}{10000+\omega^2}, && (33) \end{aligned}$$

where the final step is optional, but rationalizes the denominator, which is often the preferred form as it leads to an immediate representation of imaginary and real parts. This means the complex coordinates are

$$c = \frac{100}{10000+\omega^2} + \frac{-\omega}{10000+\omega^2} j \tag{34}$$

Answer: This is also called the Fourier transform of x.

Question 18. A time-discrete signal $s(n)$ consists of the following sequence: $1, -2, 2, 1$ for $n = [0, 3]$.

(a) Find the complex coefficients $X(e^{j\Omega})$ with the Hilbert space

$$V : \Phi = \left\{ \varphi(n) = e^{j\Omega n} \mid \Omega \in \mathbb{R}, n \in \mathbb{N} \right\}.$$

(b) What is another name for this representation?

Answer: Similarly to question 17.1, we need to find the inner product

$$\begin{aligned} X(e^{j\Omega}) &= \langle s(n), e^{j\Omega n} \rangle && (35)\\ &= \sum_{k=0}^{3} s(k) e^{-j\Omega k} && (36)\\ &= 1 \cdot e^{-j\Omega \cdot 0} + (-2) \cdot e^{-j\Omega} + 2 \cdot e^{-j\Omega \cdot 2} + 1 \cdot e^{-j\Omega \cdot 3} && (37)\\ &= 1 - 2 \cdot e^{-j\Omega} + 2 \cdot e^{-j2\Omega} + e^{-j3\Omega}. && (38) \end{aligned}$$

Answer: This is also called the discrete-time Fourier transform of s.

Question 19. Choose $N = 4$ values $\Omega_k, k = 0, \ldots, N-1$, equally spaced on the unit circle: $\Omega_K = \frac{2\pi k}{N}$.

(a) Find the complex coefficients $X(k) = X(e^{j\Omega_k})$, $k = 0, ..., N-1$, with the bases

$$\Phi = \left\{ \phi(n) = e^{j\omega_k n} \mid k \in [0, N-1] \subset \mathbb{Z}, n \in \mathbb{Z} \right\}.$$

for the same signal as in 1.18

(b) What is another name for this representation?

Answer: Let us calculate

$$\begin{aligned}
X(e^{j\Omega_k}) &= \sum_{n=0}^{3} s(n) \cdot e^{j\Omega_k n} && (39)\\
&= \sum_{n=0}^{3} s(n) \cdot e^{j\frac{2\pi kn}{N}} && (40)\\
&= 1 - 2 \cdot e^{-j\Omega_k} + 2 \cdot e^{-j2\Omega_k} + e^{-j3\Omega_k} && (41)\\
&= 1 - 2 \cdot e^{-j\frac{2\pi k}{4}} + 2 \cdot e^{-j\frac{4\pi k}{4}} + e^{-j\frac{6\pi k}{4}}, && (42)
\end{aligned}$$

for which we used Answer 18.1. This leads us to

$$\begin{aligned}
X(e^{j\Omega_0}) &= 2 && (43)\\
X(e^{j\Omega_1}) &= 1 - 2 \cdot e^{-j\frac{2\pi}{4}} + 2 \cdot e^{-j\frac{4\pi}{4}} + e^{-j\frac{6\pi}{4}} && (44)\\
&= 1 - 2e^{-j\frac{\pi}{2}} + 2e^{-j\pi} + e^{-j\frac{3\pi}{2}} && (45)\\
&= 1 - 2(-j) + 2(-1) + 1(j) && (46)\\
&= -1 + j3 && (47)\\
X(e^{j\Omega_2}) &= 1 - 2 \cdot e^{-j\frac{4\pi}{4}} + 2 \cdot e^{-j\frac{8\pi}{4}} + e^{-j\frac{12\pi}{4}} && (48)\\
&= 1 - 2 \cdot e^{-j\pi} + 2 \cdot e^{-j2\pi} + e^{-j3\pi} && (49)\\
&= 1 - 2(-1) + 2(1) + 1(-1) && (50)\\
&= 4 && (51)\\
X(e^{j\Omega_3}) &= 1 - 2 \cdot e^{-j\frac{6\pi}{4}} + 2 \cdot e^{-j\frac{12\pi}{4}} + e^{-j\frac{18\pi}{4}} && (52)\\
&= 1 - 2(j) + 2(-1) + 1(-j) && (53)\\
&= -1 - 3j. && (54)
\end{aligned}$$

Answer: This is often referred to as Discrete Fourier transform.